
Personality and Political Orientation in Germany: A NLP-based Analysis using Twitter Data

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Abstract

Despite having found some clear tendencies, research about personality and political orientation has been rather fragmented. We compliment the picture by examining systematic personality differences among sympathisers of Germany's seven main political parties as well as the relationship between personality traits and political views on economic and social issues. While previous research has only covered personality traits included in the *Big Five* model, we extend this by also covering fundamental needs and values. A natural language processing approach in combination with machine learning is applied in order to generate personality profiles based on politicians' public Twitter posts. We identify various personality aspects significantly related to certain political views, including Openness, Conscientiousness and Agreeableness as well as the needs Curiosity, Excitement and Love and the values Conservation, Hedonism and Self-transcendence. Furthermore, we find remarkable similarities between members of the AfD and Die Linke, despite their major ideological differences.

1 Introduction

The current state of research suggests that the development of political views and behaviour is the result of a complex interaction between numerous factors with varying stability. Among others, those seem to include genetic predispositions and their manifestation in the size and activity of certain brain areas and a person's intelligence, the socio-economic background of a person (education, wealth etc.), their social environment (and the views of friends, relatives and authorities), personal experiences as well as the cultural and ideological history of the country a person lives or grew up in. In addition, a number of research efforts have shown that psychological characteristics (e.g. personality traits), which are the focus of this work, are also tightly related to the formation of political views (see section 2).

Despite the fact that the idea that certain personality traits may have a strong influence on our political orientation reaches back to the 1920s (Allport (1929)), the research in this field has been fragmented and, depending on the study setup, yielded somewhat inconsistent results. While many correlations seem clear and stable, however, results about the direct and indirect causal interactions from genetic predispositions over personality traits to political orientation are mixed and remain unclear. It has to be stressed that this work, with its current approach, does not and cannot provide information about any causal effects, neither about their direction nor their strength, between genetic predispositions, personality and political views. Thus, such direct

causality, specifically between a person’s personality traits, values and needs and their political orientation, is no assumption of this work.

However, it provides valuable up-to-date insights into the fundamental psychological characteristics of the representatives and thus sympathisers of Germany’s seven largest and main political parties as well as the differences among them. By allocating those parties into a two-dimensional political landscape it also gives indications about the correlational relationship among peoples’ psychological traits and their overall political views (i.e. economically left and right, socially progressive and conservative). Furthermore, while the relationship between personality traits and political views has been examined to some extent in the past using the *Big Five* model (see section 2), we extend the analysis by also incorporating a person’s values and needs. This work thereby benefits from a novel approach of creating psychological profiles *en masse* using a machine learning and natural language processing (NLP)-based analysis of politicians’ public social media contributions, specifically the vocabulary and tonality they use therein. A detailed explanation of the approach can be found in section 3.

2 Literature Review

Different behavioural-genetic studies have consistently demonstrated that differences in peoples’ political views and behaviour are influenced by their genes. In fact, such genetic factors seem to explain 20% (Alford, Funk, & Hibbing (2005)) to 70% (Fowler & Schreiber (2008)) of those differences. Furthermore, personality traits have been shown to be highly heritable tendencies in the way people think, feel and act (Riemann & Spinath (2005)). As C Kandler & Riemann (2013) point out, there is a number of arguments suggesting that personality traits serve as a mediator between genetic factors and political views: (1) the same relationships between personality traits and general political orientation can be found in countries with different ideological backgrounds, e.g. in Germany, the US and Japan (C. Kandler et al. (2013)), (2) different longitudinal studies have shown that personality traits tend to influence attitudes more than vice versa (Sibley & Duckitt (2013)) and (3) the relationships between political orientation and personality traits can to some extent be traced back to genetic factors (Kandler, Bleidorn, & Riemann (2012)). However, while the results of those studies are in favor of a mediating effect of personality traits on political views, the results of Verhulst, Eaves, & Hatemi (2011) cast doubt on this theory. They suggest that political views are merely systematically linked to personality but not a result of them. Hence, it can be stated that there is still no consensus about the presence of causality in the described relationships nor its direction.

Nevertheless, the correlations between different personality traits and political views have been examined and documented quite well. Many studies made use of the *Big Five* approach as a standard to examine and quantify personality traits, which has become popular within the psychological field of trait theory in the 1990s (Angleitner & Ostendorf (1994)) and will be described in more detail in section 3.3. Schumann (2002) showed, for instance, that attitudes towards different political parties and movements correlate with the presence and strength of certain personality characteristics, in particular a person’s Openness to Experience, Conscientiousness and Agreeableness, exhibiting remarkable temporal stability. More specifically, Sibley, Osborne, & Duckitt (2012) found in their meta-analysis that Openness is significantly negatively and Conscientiousness positively correlated with political conservatism. Kandler et al. (2012) had similar findings showing that Openness to Experience is related to a lower resistance to change and lower acceptance of social inequality while Conscientiousness is positively correlated with resistance to change and Agreeableness negatively with the acceptance of inequality. Furthermore, Furnham & Fenton-O’Creevy (2018) showed more recently for a large British adult sample that Open, more Agreeable people tend to be more Left-Wing and Introverted, more Conscientious people more Right-Wing.

What is new about our analysis is a NLP-based approach of collecting and generating psychological profiles in large quantities and a less time-consuming way in order to examine the mentioned

relationships of interest (detailed description in section 3). Human language can provide very detailed insights into a person’s personality, their thinking style as well as emotional state. Several studies indicate that the frequency with which someone uses certain words or categories of words serves as a good predictor of their personality characteristics (Fast & Funder (2008), Hirsh & Peterson (2009)) and have applied this finding to textual data and social media contributions, such as Twitter posts and blog articles, with promising results (Gill, Scott, & Oberlander (2009), Golbeck, Robles, Edmondson, & Turner (2011), Yarkoni (2010)).

3 Methodology

In this section we describe our approach, the data used for our work as well as the analysis process in detail. We explain the three different, widely-used psychological models, the (1) *Big Five*, (2) the *Needs* and (3) *Values* model, including the interpretation of their results. Each of them will later be applied to our data in section 4 in order to investigate potential systematic personality differences among members of the various German political parties and the two major dimensions of political orientation.

3.1 Data Set

In order to create personality profiles in a standardized and easily scalable way without the need of personality tests, we used textual data from politicians’ own public Twitter posts (i.e. no retweets). For each of the seven major German parties (AfD, CDU, CSU, FDP, Die Grüne, Die Linke and SPD) we collected all public twitter posts of all their members in the German national parliament (i.e. Bundestag) and all 16 state-level parliaments (i.e. Landtage) who had a Twitter account. The individual posts were then aggregated for each politician into one text.

Subsequently, to convert the textual data into psychological profiles, we made use of an external service from IBM Watson called *Personality Insights* which “applies linguistic analytics and personality theory to infer attributes from a person’s unstructured text” (IBM (2020)). The algorithm uses an open-vocabulary approach representing a state-of-the-art method in the research about personality inference (Arnoux et al. (2017), Schwartz et al. (2013) and Plank & Hovy (2015)). In the first step, the words of the respective text are transformed into their vector representation in an n -dimensional feature space using the word-embedding technique *GloVe* (Pennington, Socher, & Manning (2014)). This vector then serves as the input for a machine learning model which, based on having been trained on thousands of Twitter users whose personality traits were known from surveys, outputs a score between 0 and 1 for each aspect of the *Big Five*, *Needs* and *Values* model, respectively. According to the documentation, one can expect a MAE of the model output of 0.12, 0.12, 0.11 and 0.11 for *Big Five* dimensions, *Big Five* facets, *Needs* and *Values* scores, respectively.

Because the *Personality Insights* model only reaches acceptable accuracy with ≥ 600 (maximum accuracy with ≥ 3000 words), we discarded all politicians with a total word count below that. Table 3.1 summarizes the resulting base data, presenting the number of politicians for each party for who we created a personality profile, their median number of tweets and their median number of total words (aggregated over all tweets).

Table 3.1: Summary - Twitter Data

	AfD	CDU	CSU	FDP	Grüne	Linke	SPD
$N_{\text{Politicians}}$	152	225	48	125	191	115	285
$\tilde{N}_{\text{Tweets}}$	86	75	95	78	62	55	73
$\tilde{N}_{\text{Words}}$	2754	2215	2294	2924	1964	2355	1979

3.2 Political Landscape and Mapping of Parties

Since one objective of this work is to not just highlight systematic personality differences among sympathisers of the different parties but also to show personality-related tendencies in the overall dimensions of political orientation, it is necessary to map the individual parties on some form of political landscape in order to quantify the ideology of each party and later use the resulting score in the regression setting described in section 3.5.

For this purpose, we built on the work of Switek, Thomeczek, & Krouwel (2017) who, before the last election of the German Bundestag, placed all seven parties into a 2-dimensional matrix depending on (1) how economically left or right they are and (2) how socially progressive or conservative they are, based e.g. on their core values, goals and agendas. The resulting mapping is presented in Figure 3.1 where all parties are colored according to their identification colors and the two dimensions are scaled to the interval $[-1, 1]$. The same color coding will also be used in subsequent figures.

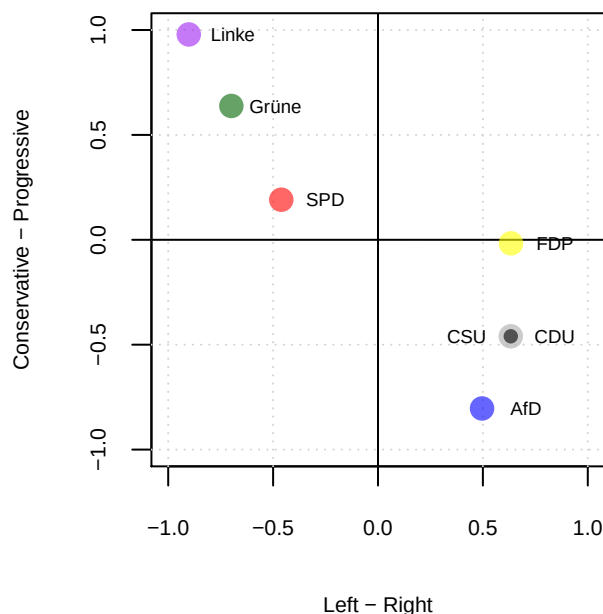


Figure 3.1: Mapping of Germany’s political landscape before the most recent election of the Bundestag (2017).

3.3 Psychological Models

3.3.1 Big Five Personality Model

The *Big Five* model developed by Costa & McCrae (1992) and Norman (1963) is the most commonly applied model to describe peoples’ personalities in a consistent and easily comparable way. It describes how a person engages with the world in general based on five dimensions, each with six more detailed facets. For a description of all facets and dimensions, please refer to Table .1 in Appendix A.

The *Personality Insights* model outputs a score for each dimension and facet between 0 and 1. According to IBM (2020), a score above 0.5 indicates a greater than average tendency for the respective characteristic while, for instance, scores of 0.25 and 0.75 can be considered *low* and *high*, respectively. This is also true for the outputs referring to the *Needs* and *Values* model.

3.3.2 Needs Model

The *Needs* model consists of 12 fundamental categories of needs which are thought to play an essential role in human behaviour. It is based on the work of Kotler & Armstrong. (2013) and Ford (2005) suggesting that a number of types of human needs are universally valid and tend to directly influence a person’s behavior. Due to a lack of space, the 12 needs and their interpretations are listed in Table .2 in Appendix A.

3.3.3 Values Model

The *Values* model was developed by Schwartz (2006) who describes them as “desirable, trans-situational goals, varying in importance, that serve as guiding principles in people’s lives” and summarizes five features which all of those values have in common:

- Values are beliefs.
- Values are a motivational construct.
- Values transcend specific actions and situations.
- Values guide the selection or evaluation of actions, policies, people, and events.
- Values vary by relative importance and can be ranked by importance.

The proposed five basic human values and their explanations are presented in Table .3 in Appendix A.

3.4 Visualization Procedure: Dimensionality Reduction

All three personality models which are applied in this work describe certain aspects of a person’s personality using multiple dimensions, specifically, the *Big Five* model in its highest level of detail uses 30 facets, the *Needs* model 12 and the *Values* model five dimensions. In order to visualize the differences, i.e. the distance between (or similarity of) multiple personality profiles, in those higher-dimensional spaces, we make use of a novel dimension reduction technique called UMAP (Uniform Manifold Approximation and Projection), originally developed by McInnes, Healy, & Melville (2018). In contrast to many other dimension reduction methods, e.g. t-SNE, UMAP preserves global data structure, i.e. not only within-cluster distances but also between-cluster similarities exhibited by the entire data set are still present when moving from high- to low-dimensional space. For each of the three personality models of interest we reduce the personality differences represented by their components to two dimensions. This allows us to plot each personality profile (each politician) in a 2-D graph together with the within-party medians. Since UMAP preserves global distances, the results can be interpreted as follows:

- politicians (represented by the small points in the graph) that are closer to each other are more similar with regard to the respective personality traits and vice versa
- within-party medians (represented by the large points) that are closer to each other indicate that their members are on average more similar with regard to the respective personality traits and vice versa

3.5 Regression Procedure

In order to answer the question “What are the personality traits that distinguish the members (or sympathisers) of one specific party from all others?”, we run a logistic one-vs-all regression for each of the seven parties and each of the four sets of personality traits. The latter include (1) all five dimensions of the *Big Five* model, (2) all 30 facets of the *Big Five* model, (3) all 12 dimensions of the *Needs* model and (4) all five dimensions of the *Values* model. Hence, for this purpose, 28 logit models will be estimated in total using maximum likelihood estimation (MLE). Consequently, for

party p , individual i and personality trait set m with D_m dimensions we define the following model specification:

$$P(Y_i = 1) = \frac{e^{\beta_0 + \sum_{d=1}^{D_m} \beta_d X_{id}}}{1 + e^{\beta_0 + \sum_{d=1}^{D_m} \beta_d X_{id}}}$$

where $Y \in \{0, 1\}$ indicates whether or not individual i is a member of party p .

Furthermore, we try to examine how a person's personality traits are related to (1) how economically left or right and (2) how socially progressive or conservative their political views are where we use the scores derived in section 3.2 for the different parties as a proxy for all individuals belonging to the respective party for model estimation. For this purpose, we will estimate 10 linear models using ordinary least squares (OLS), one for each of the four sets of personality traits (and a combination of them) and each of the two mentioned political dimensions. Hence, the following model specifications can be defined for political dimension (1), i.e. the attitude towards economic issues E and political dimension (2), i.e. the attitude towards social issues S , respectively:

$$E_i = \beta_0 + \sum_{d=1}^{D_m} \beta_d X_{id}$$

$$S_i = \beta_0 + \sum_{d=1}^{D_m} \beta_d X_{id}$$

where E_i and S_i are the scores shown in section 3.2 for party p of individual i .

4 Results

4.1 Big Five Model

The results of the *Big Five* model reduced to two dimensions are presented in Figure 4.1 and the corresponding distances of within-party medians in Table 4.1 (normalized for easier comparison). Because the five dimensions of the *Big Five* model are derived from the 30 facets, only the latter were used to create the UMAP visualization and distance table as showing both individually would be redundant. Although there is considerable overlap among the personality traits of the individual politicians, the within-party medians reveal interesting similarities among the parties. For instance, members of the AfD seem (on average) to be most similar to members of Die Linke with regard to the traits comprised by the *Big Five* model. Furthermore, members of the CDU seem to be most similar to those of the SPD while members of Die Grünen are closest to those of the FDP.

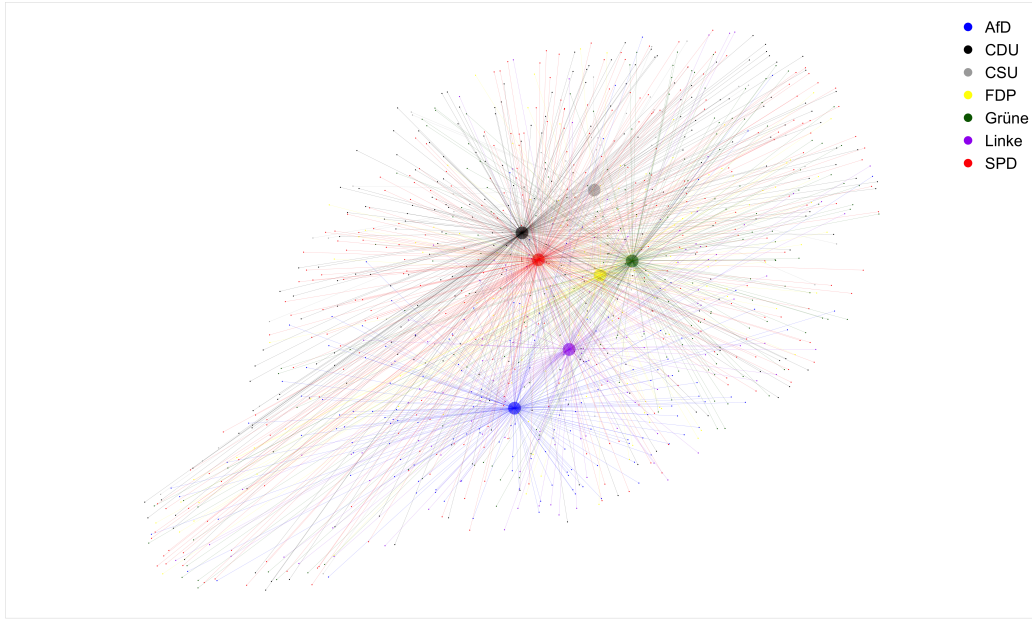


Figure 4.1: 2-dimensional UMAP of Big Five personality traits (all 30 facets) with highlighted within-group medians.

Table 4.1: Normalized distances of within-party medians.

	AfD	CDU	CSU	FDP	Grüne	Linke	SPD
AfD	0.00	0.79	1.00	0.64	0.73	0.30	0.67
CDU	0.79	0.00	0.27	0.28	0.32	0.54	0.13
CSU	1.00	0.27	0.00	0.38	0.33	0.72	0.35
FDP	0.64	0.28	0.38	0.00	0.11	0.34	0.18
Grüne	0.73	0.32	0.33	0.11	0.00	0.43	0.25
Linke	0.30	0.54	0.72	0.34	0.43	0.00	0.41
SPD	0.67	0.13	0.35	0.18	0.25	0.41	0.00

The detailed results of the one-vs-all logistic regressions concerning the *Big Five* dimensions and facets can, due to a lack of space, be found in Table .4 and .5 in Appendix B, respectively. However, we plotted the distribution of scores for each *Big Five* dimension and each party in Figure 4.2 and highlighted those which the respective regression found to be a significantly characteristic personality trait of members of a certain party. Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all logistic regression. In addition, the overall median for each dimension is presented. For instance, the figure shows with regard to Openness that members of the CSU are characteristically less and those of Die Grünen more open than others as the corresponding coefficients in the one-vs-all regression are both significant at the 5% confidence level. The rest can be interpreted accordingly. A similar visualization for all the 30 facets can be found in Appendix C.

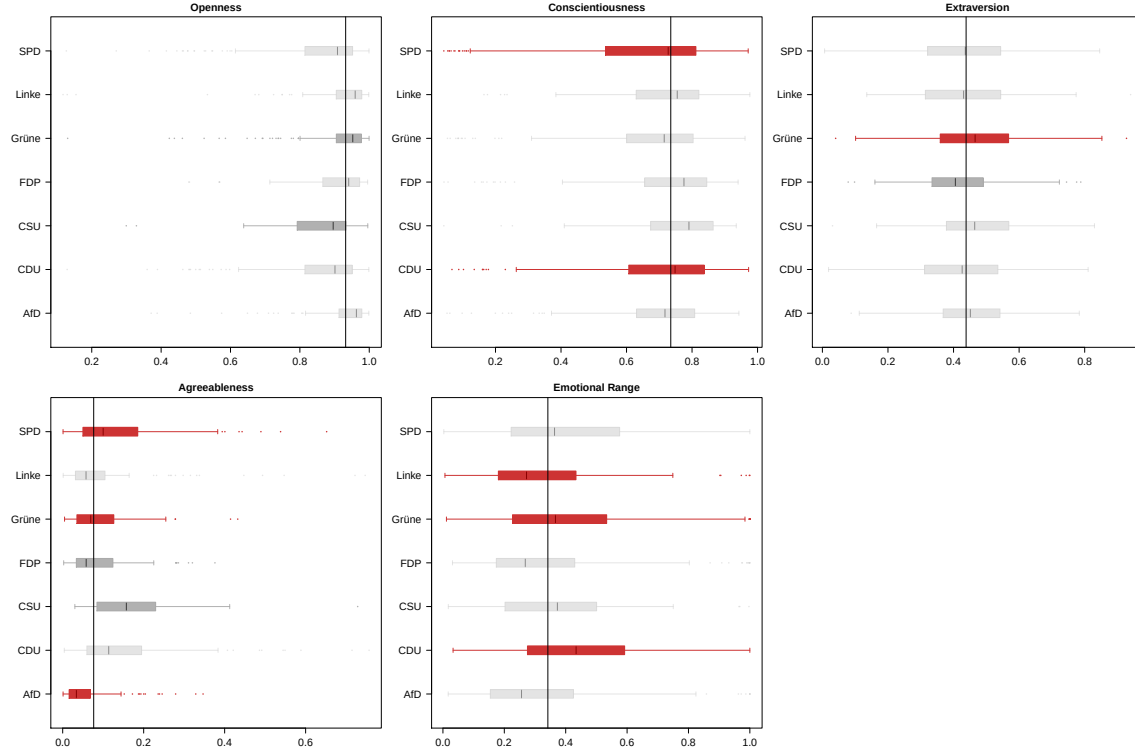


Figure 4.2: Distributions of scores for each party and each personality trait (5 primary dimensions). Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all regression.

4.2 Needs Model

The results of the *Needs* model reduced to two dimensions are presented in Figure 4.3 and the corresponding distances of within-party medians in Table 4.2. It is remarkable how much different the needs of AfD members appear to be since the within-party median is noticeably isolated from those of the other parties. Nevertheless, it is again closest to the median of Die Linke and second closest to the median of Die Grünen despite their obviously major ideological differences. Furthermore, although CDU and CSU are sister parties, the needs of their members appear to be relatively dissimilar. However, please note that the sample size for the CSU is comparably small, potentially leading to higher-variance-induced deviations.

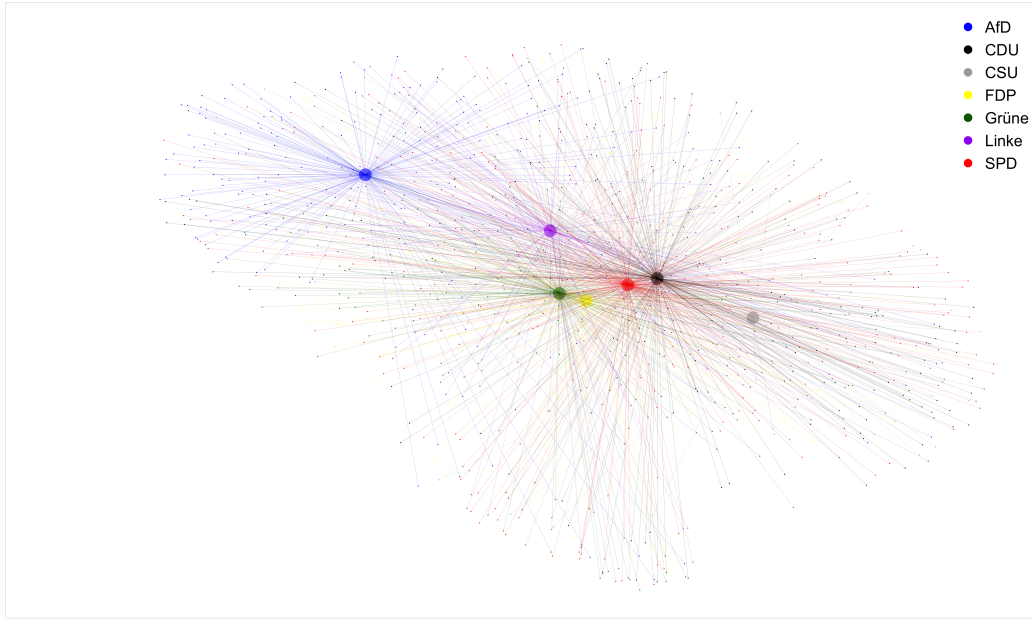


Figure 4.3: 2-dimensional UMAP of 12 needs with highlighted within-group medians.

Table 4.2: Normalized distances of within-party medians.

	AfD	CDU	CSU	FDP	Grüne	Linke	SPD
AfD	0.00	0.75	1.00	0.67	0.61	0.45	0.70
CDU	0.75	0.00	0.26	0.18	0.22	0.29	0.07
CSU	1.00	0.26	0.00	0.37	0.43	0.55	0.30
FDP	0.67	0.18	0.37	0.00	0.06	0.27	0.11
Grüne	0.61	0.22	0.43	0.06	0.00	0.23	0.15
Linke	0.45	0.29	0.55	0.27	0.23	0.00	0.26
SPD	0.70	0.07	0.30	0.11	0.15	0.26	0.00

Again, the detailed regression results can be found in Table .6 in Appendix B while the distributions of the individual needs are presented for each party in Figure 4.4 and the significance of the corresponding coefficients in the regression is highlighted.

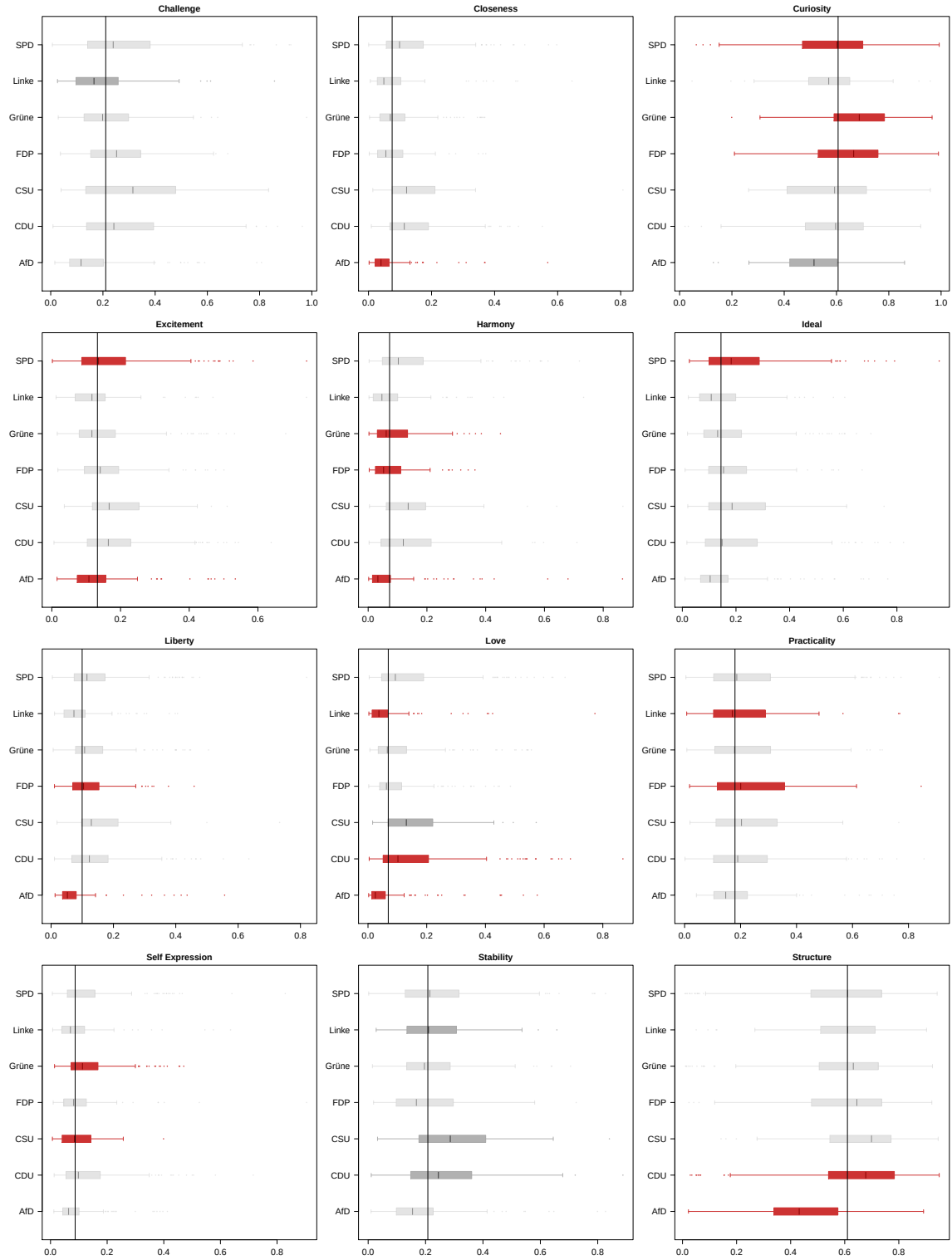


Figure 4.4: Distributions of scores for each party and each need. Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all regression.

4.3 Values Model

The results of the *Values* model reduced to two dimensions are presented in Figure 4.5 and the corresponding distances of within-party medians in Table 4.3. The relatively high similarity of members of Die Linke and AfD also appears to be present with regard to the examined five fundamental values. Moreover, the median of members of Die Grünen is somewhat isolated in the graph while members of the CDU and SPD seem to have very similar values.

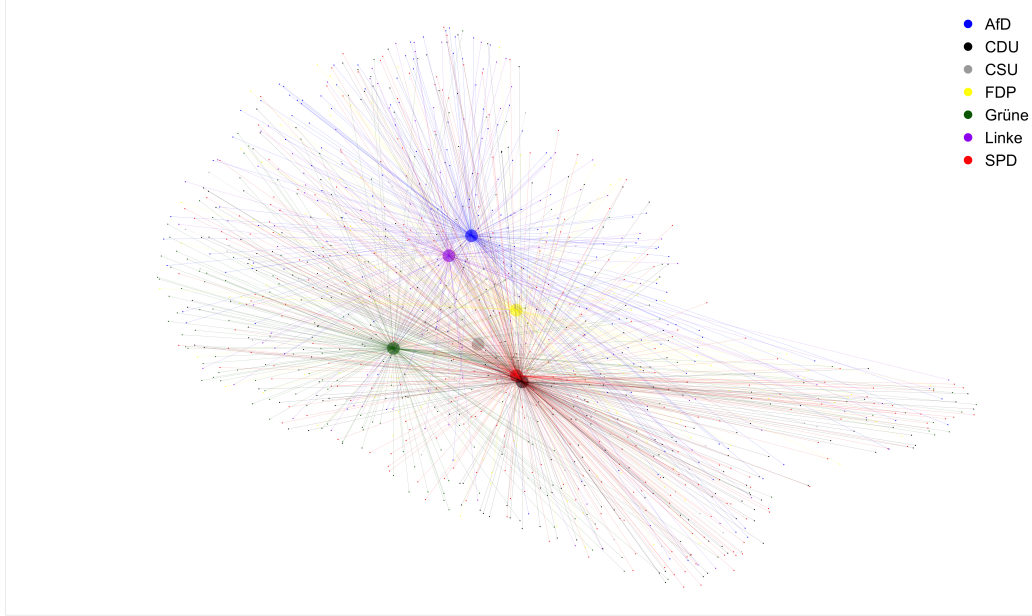


Figure 4.5: 2-dimensional UMAP of five values with highlighted within-group medians.

Table 4.3: Normalized distances of within-party medians.

	AfD	CDU	CSU	FDP	Grüne	Linke	SPD
AfD	0.00	1.00	0.73	0.53	0.82	0.16	0.95
CDU	1.00	0.00	0.31	0.48	0.56	0.90	0.05
CSU	0.73	0.31	0.00	0.28	0.34	0.61	0.26
FDP	0.53	0.48	0.28	0.00	0.55	0.45	0.44
Grüne	0.82	0.56	0.34	0.55	0.00	0.66	0.52
Linke	0.16	0.90	0.61	0.45	0.66	0.00	0.85
SPD	0.95	0.05	0.26	0.44	0.52	0.85	0.00

The distributions of the individual values are presented for each party in Figure 4.6 and the significance of the corresponding coefficients in the regression are highlighted as explained. The original regression results can be found in Table .7 in Appendix B.

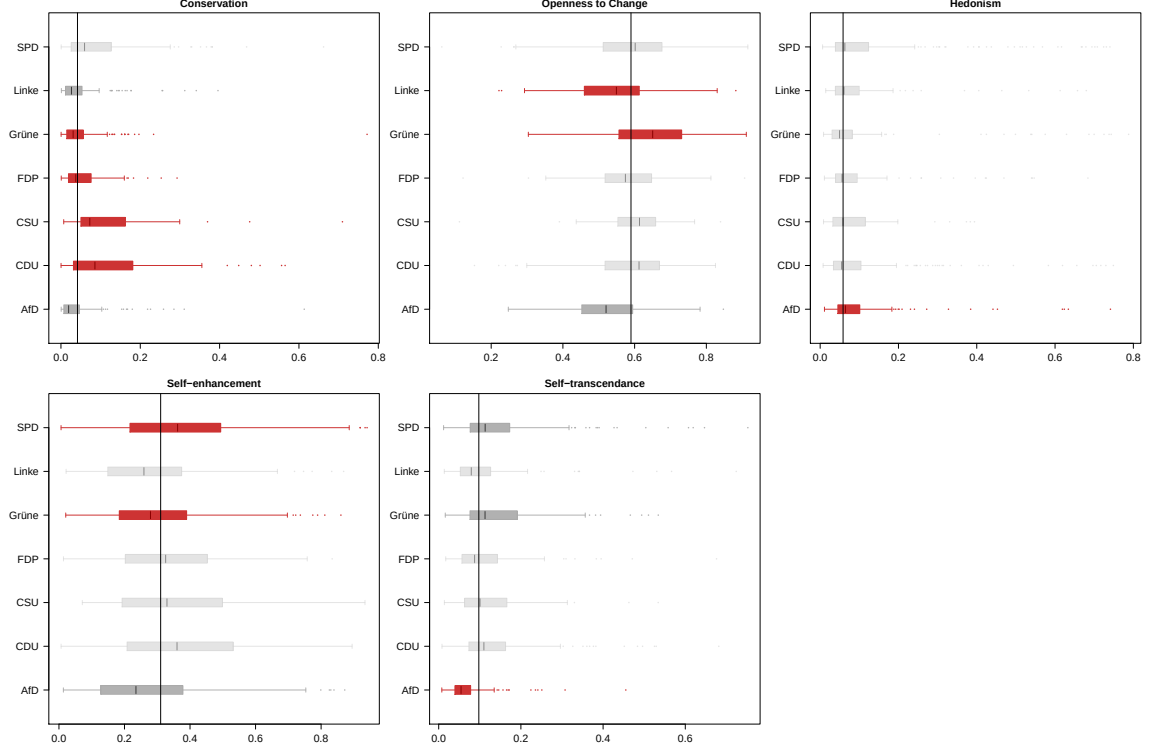


Figure 4.6: Distributions of scores for each party and each value. Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all regression.

4.4 Personality Tendencies among General Political Dimensions

In order to be able to make specific statements about the relationship between certain personality traits and having left or right as well as progressive or conservative political views, we conducted multiple linear regressions. For this purpose, we defined the dependent variable E to represent a person's attitude towards economic issues where $E \in [-1, 1]$ and higher (lower) values indicate a more Right-Wing (Left-Wing) view and the dependent variable S to represent a person's attitude towards social issues where $S \in [-1, 1]$ and higher (lower) values indicate a more progressive (conservative) view as presented in section 3.2. Specifically, the following models were estimated where the results of model 1-8 are shown in Table 4.4 and those of model 9 and 10, due to lack of space, in Table .8 in Appendix D:

1. dep. variable: E - 'being Right-Wing'; indep. variables: *Big Five* model dimensions
2. dep. variable: S - 'being progressive'; indep. variables: *Big Five* model dimensions
3. dep. variable: E - 'being Right-Wing'; indep. variables: *Needs* model dimensions
4. dep. variable: S - 'being progressive'; indep. variables: *Needs* model dimensions
5. dep. variable: E - 'being Right-Wing'; indep. variables: *Values* model dimensions
6. dep. variable: S - 'being progressive'; indep. variables: *Values* model dimensions
7. dep. variable: E - 'being Right-Wing'; indep. variables: *Big Five* + *Needs* + *Values* dimensions
8. dep. variable: S - 'being progressive'; indep. variables: *Big Five* + *Needs* + *Values* dimensions
9. dep. variable: E - 'being Right-Wing'; indep. variables: *Big Five* facets
10. dep. variable: S - 'being progressive'; indep. variables: *Big Five* facets

We find that the best fitting models, even after adjusting for the additional degrees of freedom used, are model 9 and 10 (Adj. $R^2 = 0.135$ and 0.190) just before model 7 and 8 (Adj. $R^2 = 0.085$

and 0.125). Based on model 7 and 8 and only taking coefficients into account with $p < 0.05$, we can make the following inferences:

Table 4.4: Linear regression results (model 1-8) with genereal political dimensions as target variables.

	<i>Dependent variable:</i>							
	E	S	E	S	E	S	E	S
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Big Five: Openness	-0.277 (0.184)	0.282* (0.165)					0.889*** (0.319)	-1.153*** (0.278)
Big Five: Conscientiousness	0.372*** (0.121)	-0.183* (0.108)					0.758*** (0.223)	-0.576*** (0.195)
Big Five: Extraversion	-0.077 (0.130)	-0.016 (0.117)					-0.278* (0.160)	0.190 (0.140)
Big Five: Agreeableness	-0.117 (0.208)	0.124 (0.186)					-0.497 (0.345)	0.886*** (0.302)
Big Five: Emotional Range	0.108 (0.106)	-0.036 (0.095)					0.065 (0.122)	-0.040 (0.107)
Needs: Challenge			0.223 (0.178)	-0.176 (0.156)			0.191 (0.246)	-0.377* (0.215)
Needs: Closeness			-0.533 (0.392)	1.027*** (0.343)			0.102 (0.513)	0.061 (0.448)
Needs: Curiosity			-0.057 (0.152)	0.389*** (0.134)			-0.002 (0.187)	0.484*** (0.163)
Needs: Excitement			1.147*** (0.370)	-0.875*** (0.324)			1.488*** (0.455)	-0.981** (0.397)
Needs: Harmony			0.747** (0.328)	-1.413*** (0.287)			-0.164 (0.401)	-0.791** (0.350)
Needs: Ideal			-0.408 (0.278)	0.303 (0.244)			-0.122 (0.347)	0.163 (0.303)
Needs: Liberty			-0.897* (0.476)	0.708* (0.417)			-0.287 (0.621)	-0.045 (0.543)
Needs: Love			0.526* (0.300)	-0.212 (0.263)			1.386*** (0.338)	-0.920*** (0.295)
Needs: Practicality			-0.096 (0.215)	0.350* (0.188)			-0.167 (0.257)	0.481** (0.224)
Needs: Self Expression			-0.800** (0.336)	0.593** (0.295)			-0.229 (0.364)	0.019 (0.318)
Needs: Stability			0.018 (0.209)	-0.058 (0.183)			-0.244 (0.213)	0.151 (0.186)
Needs: Structure			-0.215 (0.157)	0.480*** (0.137)			-0.914*** (0.188)	1.087*** (0.164)
Values: Conservation					0.985*** (0.239)	-0.939*** (0.212)	1.754*** (0.414)	-1.706*** (0.361)
Values: Openness to Change					-0.386** (0.173)	0.378** (0.154)	-0.445* (0.236)	0.263 (0.206)
Values: Hedonism					0.024 (0.170)	-0.175 (0.151)	-0.817** (0.347)	0.387 (0.303)
Values: Self-enhancement					0.107 (0.124)	-0.046 (0.110)	-0.056 (0.192)	0.228 (0.168)
Values: Self-transcendence					-0.733*** (0.230)	0.869*** (0.204)	-0.614* (0.340)	0.819*** (0.297)
Constant	-0.040 (0.192)	-0.084 (0.172)	0.093 (0.107)	-0.492*** (0.094)	0.169* (0.091)	-0.191** (0.081)	-0.611** (0.300)	0.397 (0.262)
Observations	1,141	1,141	1,141	1,141	1,141	1,141	1,141	1,141
R ²	0.010	0.004	0.032	0.068	0.037	0.047	0.103	0.142
Adjusted R ²	0.006	-0.00004	0.022	0.058	0.033	0.043	0.085	0.125

Note:

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

- Openness seems related to being Right-Wing and conservative ($p < 0.01$).
- Conscientiousness seems related to being Right-Wing and conservative ($p < 0.01$).
- Agreeableness seems related to being progressive ($p < 0.01$).
- The need Curiosity seems related to being progressive ($p < 0.01$).
- The need Excitement seems related to being Right-Wing ($p < 0.01$) and conservative ($p < 0.05$).
- The need Harmony seems related to being conservative ($p < 0.05$).
- The need Love seems related to being Right-Wing and conservative ($p < 0.01$).
- The need Practicality seems related to being progressive ($p < 0.05$).
- The need Structure seems related to being Left-Wing and progressive ($p < 0.05$).
- The value Conservation seems related to being Right-Wing and conservative ($p < 0.01$).

- The value Hedonism seems related to being Left-Wing ($p < 0.05$).
- The value Self-transcendence seems related to being progressive ($p < 0.01$).

Furthermore, according to the results of model 9 and 10 in Appendix D, the following statements can be made:

- Artistic Interests seem related to being Right-Wing and conservative ($p < 0.01$).
- Emotionality seems related to being Left-Wing and progressive ($p < 0.01$).
- Imagination seems related to being Left-Wing ($p < 0.05$) and progressive ($p < 0.01$).
- Intellect seems related to being Right-Wing and conservative ($p < 0.01$).
- Challenging Authority seems related to being Left-Wing and progressive ($p < 0.01$).
- Cautiousness seems related to being Right-Wing ($p < 0.01$) and conservative ($p < 0.01$).
- Self-discipline seems related to being Left-Wing ($p < 0.05$) and progressive ($p < 0.01$).
- Being Energetic seems related to being progressive ($p < 0.05$).
- Assertiveness seems related to being Right-Wing and conservative ($p < 0.01$).
- Seeking Excitement seems related to being Right-Wing and conservative ($p < 0.01$).
- Altruism seems related to being conservative ($p < 0.01$).
- Modesty seems related to being Left-Wing ($p < 0.01$) and progressive ($p < 0.05$).
- Being Uncompromising seems related to being Right-Wing ($p < 0.05$) and conservative ($p < 0.01$).
- Sympathy seems related to being Left-Wing and progressive ($p < 0.01$).
- Being Fiery seems related to being Right-Wing and conservative ($p < 0.01$).
- Being Prone to Worry seems related to being conservative ($p < 0.05$).
- Self-consciousness seems related to being Left-Wing and progressive ($p < 0.05$).

5 Conclusion

Collecting conversational social media data from Twitter and generating personality profiles based on this textual data using machine learning has proven to be a very convenient and easily scalable approach without the need to conduct and evaluate numerous surveys / personality tests. We have shown that there appears to be strong evidence for systematic differences in personality traits (according to the *Big Five* model), needs and values among the members of the seven main political parties in Germany (despite considerable overlap). Quantifying their parties' political views on economic and social issues allowed us to reveal similar systematic personality tendencies with regard to how Left-Wing or Right-Wing and progressive or conservative a person is. This suggests that a collection of personality aspects could serve as a good predictor for how a person will vote and which parties they would sympathize with and which not.

Complementing the results of Schumann (2002), we also found the *Big Five* dimensions Openness, Conscientiousness and Agreeableness to be highly related to person's political views. We extended the personality aspects included in the *Big Five* model by the components of the *Values* and *Needs* model, which - to our knowledge - have not been covered in this context by previous research efforts. It appears that some fundamental values and needs are also highly related to political orientation, including Curiosity, Excitement and Love for the former and Conservation, Hedonism and Self-transcendence for the latter. Nevertheless, some of our findings have shown to be contradictory to the results of other works. Sibley et al. (2012), Kandler et al. (2012) as well as Furnham & Fenton-O'Creevy (2018) all found (although partly using slightly different political dimensions) that Openness is generally positively correlated with being politically Left-Wing / progressive where we found the complete opposite to be the case for our data. Apart from that, however, our findings with regard to the sign of the political orientation-related correlations of Conscientiousness and Agreeableness are congruent with the results of those authors. Another remarkable finding is that, despite occupying the political spectrum in Germany on opposite ends, the members of the AfD and Die Linke appear to have relatively similar personality traits, needs and values. It would be quite interesting to have this verified, perhaps using different methods, by future research.

We identified several shortcomings or potential sources of distortions in our approach which might have caused slightly deviating results from the ones previously mentioned. For instance, generating personality profiles from social media posts might be error-prone due to an insufficient word count, necessary translations or a low personality-related information content of the posts and might not reflect the same results one would potentially obtain using personality surveys. Furthermore, assuming the entire parties' view on economic and social issues (i.e. its position in the political landscape) for all their members and hence using the same dependent variable value for all members of a party in our linear regression is inaccurate. The political views, even of members of one party, will certainly be subject to some variation.

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Appendix A

The following descriptions of the individual model aspects are adopted from IBM (2020).

Table .1: Dimensions and facets the Big Five Model and their Interpretation

Dimension	Facet	Interpretation
Openness	Adventurousness	Extent to which a person is open to experiencing different activities. Being eager to try new activities and experience different things. Finding familiarity and routine boring.
	Artistic Interests	Loving beauty, both in art and in nature. Becoming easily involved and absorbed in artistic and natural events. With intellect, this facet is one of the two most important, central aspects of this dimension.
	Emotionality	Having good access to and awareness of one's own feelings.
	Imagination	Viewing the real world as often too plain and ordinary. Using fantasy not as an escape but as a way of creating for oneself a richer and more interesting inner-world.
	Intellect	Being intellectually curious and tending to think in symbols and abstractions.
Conscientiousness	Authority challeng- ing	Having a readiness to challenge authority, convention, and traditional values.
	Achievement striv- ing	A person's tendency to act in an organized or thoughtful way. Trying hard to achieve excellence. Drive to be recognized as successful keeps one on track as one works hard to accomplish their goals.
	Cautiousness	Being disposed to think through possibilities carefully before acting.
	Dutifulness	Having a strong sense of duty and obligation.
	Orderliness	Being well-organized, tidy, and neat.
Extraversion	Self-discipline	Having the self-discipline, or will-power, to persist at difficult or unpleasant tasks until they are completed.
	Self-efficacy	Being confident in one's ability to accomplish things.
	Energetic	A person's tendency to seek stimulation in the company of others. Leading fast-paced and busy lives. Doing things and moving about quickly, energetically, and vigorously, and being involved in many activities.
	Assertiveness	Liking to take charge and direct the activities of others. Tending to be leaders in groups.
	Cheerfulness	Experiencing a range of positive feelings, including happiness, enthusiasm, optimism, and joy.
Agreeableness	Excitement seeking	Being easily bored without high levels of stimulation.
	Outgoing	Genuinely liking other people and openly demonstrate positive feelings toward others.
	Gregariousness	Finding the company of others pleasantly stimulating and rewarding. Enjoying the excitement of crowds.
	Altruism	A person's tendency to be compassionate and cooperative toward others. Finding that helping others is genuinely rewarding, that doing things for others is a form of self-fulfillment rather than self-sacrifice.
	Cooperation	Disliking confrontation. Being perfectly willing to compromise or to deny one's own needs to get along with others.
Emotional Range	Modesty	Being unassuming, rather self-effacing, and humble. However, not necessarily lacking self-confidence or self-esteem.
	Uncompromising	Seeing no need for pretense or manipulation when dealing with others and being therefore candid, frank, and genuine.
	Sympathy	Being tender-hearted and compassionate.
	Trust	Assuming that most people are fundamentally fair, honest, and have good intentions. Taking people at face value and are willing to forgive and forget.
		The extent to which a person's emotions are sensitive to the individual's environment.
	Fiercy	Having a tendency to feel angry.
	Prone to Worry	Often feeling like something unpleasant, threatening, or dangerous is about to happen. The fight-or-flight system of one's brain is too easily and too often engaged.
	Melancholy	Tending to react more readily to life's ups and downs.
	Immoderation	Feeling strong cravings and urges that one has difficulty resisting, even though they know that they are likely to regret them later. Tending to be oriented toward short-term pleasures and rewards rather than long-term consequences.
	Self-consciousness	Being sensitive about what others think of them. One's concerns about rejection and ridicule cause them to feel shy and uncomfortable around others; Being easily embarrassed.
	Susceptible to Stress	Having difficulty coping with stress. Experiencing panic, confusion, and helplessness when under pressure or when facing emergency situations.

Table .2: Dimensions of the Needs Model and their Interpretation

Value	Interpretation
Challenge	Having an urge to achieve, to succeed, and to take on challenges.
Closeness	Relishing being connected to family and setting up a home.
Curiosity	Having a desire to discover, find out, and grow.
Excitement	Wanting to get out there and live life, having upbeat emotions, and wanting to have fun.
Harmony	Appreciating other people, their viewpoints, and their feelings.
Ideal	Desiring perfection and a sense of community.
Liberty	Having a desire for fashion and new things, as well as the need for escape.
Love	Enjoying social contact, whether one-to-one or one-to-many. Everything bringing people together taps this need.
Practicality	Having a desire to get the job done, a desire for skill and efficiency, which can include physical expression and experience.
Self-expression	Enjoying discovering and asserting their own identities.
Stability	Seeking equivalence in the physical world. They favor the sensible, the tried and tested.
Structure	Exhibiting groundedness and a desire to hold things together. The need of things to be well organized and under control.

Table .3: Dimensions of the Values Model and their Interpretation

Value	Interpretation
Conservation	Emphasizing self-restriction, order, and resistance to change.
Openness to Change	Emphasizing independent action, thought, and feeling, as well as a readiness for new experiences.
Hedonism	Seeking pleasure and sensuous gratification for themselves.
Self-enhancement	Seeking personal success for themselves.
Self-transcendence	Showing concern for the welfare and interests of others.

Appendix B

.1 Big Five Personality Traits (Primary Dimensions)

Table .4: One-vs-All Regression Results - Big Five Personality Traits (5 Primary Dimensions)

	<i>Dependent variable:</i>						
	AfD	CDU	CSU	FDP	Grüne	Linke	SPD
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Openness	0.437 (1.179)	-0.694 (0.690)	-2.394** (1.207)	0.435 (1.089)	2.654** (1.034)	0.955 (1.171)	-0.885 (0.641)
Conscientiousness	-0.251 (0.655)	2.043*** (0.495)	1.447 (1.048)	0.708 (0.674)	-0.065 (0.532)	-0.305 (0.702)	-1.356*** (0.445)
Extraversion	0.462 (0.656)	0.198 (0.534)	0.883 (1.035)	-1.620** (0.695)	1.490*** (0.578)	-0.974 (0.706)	-0.374 (0.489)
Agreeableness	-9.864*** (1.955)	1.393* (0.763)	2.891** (1.224)	-2.778** (1.390)	-3.108*** (1.149)	0.618 (1.147)	2.531*** (0.731)
Emotional Range	-0.720 (0.567)	1.973*** (0.433)	-0.636 (0.853)	-0.432 (0.580)	1.621*** (0.470)	-1.571*** (0.607)	-0.782* (0.400)
Constant	-1.252 (1.166)	-3.281*** (0.754)	-2.607* (1.375)	-1.851* (1.116)	-4.950*** (1.031)	-1.906 (1.191)	0.772 (0.685)
Observations	1,141	1,141	1,141	1,141	1,141	1,141	1,141
Log Likelihood	-411.519	-540.863	-188.600	-385.019	-497.560	-366.294	-626.329
Akaike Inf. Crit.	835.038	1,093.726	389.201	782.039	1,007.120	744.588	1,264.659

Note:

*p<0.1; **p<0.05; ***p<0.01

.2 Big Five Personality Traits (Facets)

Table .5: One-vs-All Regression Results - Big Five Personality Traits (30 Facets)

	<i>Dependent variable:</i>						
	AfD (1)	CDU (2)	CSU (3)	FDP (4)	Grüne (5)	Linke (6)	SPD (7)
Openness: Adventurousness	2.926 (2.205)	-5.372*** (1.384)	-2.574 (3.016)	-0.248 (1.827)	-0.469 (1.698)	2.351 (1.973)	-1.304 (1.230)
Openness: Artistic Interests	1.034 (2.234)	1.984 (1.271)	6.458** (2.510)	0.018 (1.663)	-1.845 (1.518)	-3.886** (1.918)	-0.504 (1.152)
Openness: Emotionality	-7.782*** (2.653)	-5.222*** (1.407)	-2.769 (3.078)	1.629 (1.798)	1.331 (1.395)	4.129** (1.787)	0.792 (1.167)
Openness: Imagination	-3.717 (2.550)	-1.082 (1.459)	-10.166*** (3.447)	5.174** (2.031)	4.334*** (1.649)	0.351 (2.050)	-0.758 (1.346)
Openness: Intellect	1.043 (4.670)	-0.401 (1.919)	6.811* (3.960)	9.055* (4.884)	-5.232* (3.131)	-13.604*** (3.908)	1.677 (1.860)
Openness: Authority challenging	-5.637* (3.157)	-0.148 (1.656)	-9.003*** (2.880)	-5.342** (2.621)	8.888*** (3.044)	13.192*** (3.411)	1.035 (1.632)
Conscientiousness: Achievement striving	-13.766*** (3.234)	0.647 (1.974)	11.357*** (3.973)	1.743 (2.604)	-5.223** (2.094)	2.943 (2.665)	3.563** (1.729)
Conscientiousness: Cautiousness	8.078*** (3.075)	3.330* (1.743)	-6.802* (3.595)	7.569*** (2.685)	-4.637** (2.066)	-10.749*** (2.741)	-1.649 (1.619)
Conscientiousness: Dutifulness	-3.947* (2.274)	1.853 (1.315)	-1.283 (2.772)	4.687*** (1.776)	-4.290*** (1.512)	2.248 (1.954)	1.022 (1.258)
Conscientiousness: Orderliness	6.601*** (2.493)	0.156 (1.405)	3.172 (2.926)	-4.121** (1.886)	1.663 (1.590)	1.330 (2.054)	-3.209** (1.295)
Conscientiousness: Self-discipline	-7.627*** (2.578)	1.608 (1.484)	-0.550 (3.294)	-1.086 (2.098)	5.941*** (1.821)	5.896** (2.385)	-2.785** (1.406)
Conscientiousness: Self-efficacy	7.101 (4.434)	-8.171*** (2.321)	-1.505 (6.157)	4.527 (3.407)	-0.426 (2.652)	-2.575 (3.270)	1.272 (2.143)
Extraversion: Energetic	-4.709 (3.621)	7.025** (2.892)	-11.854** (6.040)	-2.552 (3.165)	4.319 (3.032)	-3.175 (3.342)	3.497 (2.303)
Extraversion: Assertiveness	-0.036 (3.925)	2.399 (2.112)	1.877 (4.436)	1.731 (2.804)	-3.288 (2.371)	-0.161 (2.781)	-4.360** (1.973)
Extraversion: Cheerfulness	-5.731 (4.910)	-2.441 (1.667)	2.150 (3.328)	-12.059*** (3.275)	-2.071 (2.143)	-1.464 (3.059)	2.514 (1.606)
Extraversion: Excitement seeking	1.754 (2.423)	7.167*** (1.507)	1.965 (3.517)	0.565 (2.003)	-5.021*** (1.720)	-8.653*** (2.367)	1.110 (1.341)
Extraversion: Outgoing	-0.610 (2.923)	1.149 (1.524)	2.147 (3.107)	-0.507 (2.197)	-1.494 (1.781)	8.832*** (2.578)	2.195 (1.443)
Extraversion: Gregariousness	-1.993 (3.769)	0.309 (1.636)	-4.944 (3.400)	9.111*** (2.374)	-1.630 (2.032)	1.108 (2.867)	-5.178*** (1.630)
Agreeableness: Altruism	10.823*** (2.690)	3.373** (1.698)	-2.414 (3.591)	-5.544*** (2.037)	-1.751 (1.720)	-6.712*** (2.189)	0.289 (1.516)
Agreeableness: Cooperation	-2.639 (2.949)	3.288** (1.587)	-5.085* (3.030)	0.396 (2.102)	3.705** (1.828)	-6.765** (2.767)	-2.606* (1.443)
Agreeableness: Modesty	-0.553 (2.906)	-2.225 (1.626)	-9.638** (4.588)	-1.347 (2.419)	0.066 (1.871)	5.298** (2.214)	0.724 (1.495)
Agreeableness: Uncompromising	4.187 (2.681)	4.019*** (1.426)	3.343 (3.000)	-0.852 (1.908)	-0.858 (1.545)	-4.930** (2.307)	0.009 (1.314)
Agreeableness: Sympathy	0.292 (2.981)	-3.013** (1.425)	2.483 (3.336)	-1.824 (1.925)	4.505*** (1.662)	17.157*** (3.244)	-2.766** (1.290)
Agreeableness: Trust	1.475 (2.310)	0.520 (1.688)	11.181** (4.467)	-1.591 (1.914)	3.543** (1.795)	0.556 (2.220)	-2.533* (1.352)
Emotional Range: Fiery	0.761 (2.439)	6.542*** (1.592)	-2.867 (3.294)	0.407 (2.100)	-2.321 (1.813)	0.890 (2.308)	-4.762*** (1.429)
Emotional Range: Prone to Worry	14.745*** (3.497)	-1.293 (1.970)	-6.152 (3.769)	-5.817** (2.856)	0.254 (2.460)	-11.987*** (3.506)	2.782 (1.854)
Emotional Range: Melancholy	0.820 (2.711)	2.262 (1.623)	0.625 (3.336)	-1.840 (2.232)	-1.514 (1.879)	7.310*** (2.586)	-4.337*** (1.507)
Emotional Range: Immoderation	0.104 (3.333)	0.665 (1.795)	7.656** (3.752)	-0.040 (2.760)	-0.410 (2.074)	-6.856*** (2.281)	1.058 (1.598)
Emotional Range: Self-consciousness	-5.435** (2.667)	0.272 (1.531)	2.811 (3.574)	-3.116 (2.089)	-1.408 (1.749)	7.162*** (2.479)	2.706* (1.427)
Emotional Range: Susceptible to Stress	-10.287** (4.094)	-1.560 (2.204)	9.859** (4.543)	8.330*** (3.135)	0.547 (2.679)	7.708** (3.473)	-3.098 (2.055)
Constant	-1.464 (6.069)	-10.885*** (3.297)	-2.548 (7.025)	-11.244** (5.393)	-2.974 (4.378)	-7.693 (5.444)	5.020* (2.995)
Observations	1,141	1,141	1,141	1,141	1,141	1,141	1,141
Log Likelihood	-246.375	-487.791	-149.300	-336.678	-422.058	-259.819	-577.305
Akaike Inf. Crit.	554.750	1,037.581	360.600	735.355	906.116	581.638	1,216.610

Note:

*p<0.1; **p<0.05; ***p<0.01

.3 Needs

Table .6: One-vs-All Regression Results - 12 Needs

	<i>Dependent variable:</i>						
	AfD (1)	CDU (2)	CSU (3)	FDP (4)	Grüne (5)	Linke (6)	SPD (7)
Challenge	-2.411* (1.278)	-0.324 (0.730)	-0.565 (1.413)	1.712* (0.993)	-0.509 (0.893)	-2.752** (1.209)	0.436 (0.652)
Closeness	-12.356*** (3.479)	0.887 (1.530)	1.141 (3.120)	4.323 (2.708)	0.555 (1.979)	4.148* (2.355)	1.508 (1.461)
Curiosity	-2.334** (0.907)	-0.699 (0.648)	0.634 (1.352)	4.737*** (0.915)	4.054*** (0.743)	-1.187 (0.822)	-1.529*** (0.594)
Excitement	10.233*** (2.505)	2.842* (1.495)	-1.722 (2.751)	3.973* (2.122)	0.112 (1.833)	-2.581 (2.306)	-4.563*** (1.449)
Harmony	12.537*** (2.359)	1.385 (1.299)	-0.911 (2.408)	-11.225*** (2.728)	-5.531*** (1.865)	-3.879* (2.169)	0.662 (1.189)
Ideal	3.352* (1.884)	-1.504 (1.174)	1.281 (2.401)	0.721 (1.566)	-2.044 (1.303)	2.836 (1.793)	3.046*** (1.064)
Liberty	-21.054*** (4.372)	-0.720 (2.016)	6.070* (3.319)	-8.456*** (3.166)	3.948* (2.363)	-5.162 (3.202)	3.476* (1.789)
Love	-9.266*** (2.550)	3.912*** (1.205)	5.199** (2.136)	1.953 (1.880)	-0.032 (1.438)	-6.041*** (2.289)	0.362 (1.104)
Practicality	-0.686 (1.438)	-1.769* (0.905)	1.145 (1.777)	4.153*** (1.223)	-0.811 (0.995)	4.615*** (1.292)	-0.897 (0.829)
Self Expression	-0.677 (2.458)	1.235 (1.419)	-17.329*** (4.072)	-1.705 (2.118)	4.644*** (1.482)	1.575 (1.884)	-1.537 (1.364)
Stability	1.190 (1.270)	-2.217** (0.890)	3.803** (1.553)	-0.201 (1.233)	-1.438 (1.010)	2.545** (1.102)	-1.070 (0.819)
Structure	-6.325*** (0.954)	3.183*** (0.702)	0.806 (1.374)	0.357 (0.898)	1.225 (0.753)	0.612 (0.887)	0.488 (0.607)
Constant	3.839*** (0.596)	-2.898*** (0.477)	-4.762*** (0.992)	-5.662*** (0.698)	-4.446*** (0.570)	-1.993*** (0.593)	-0.554 (0.406)
Observations	1,141	1,141	1,141	1,141	1,141	1,141	1,141
Log Likelihood	-308.233	-528.445	-173.339	-357.758	-460.457	-345.104	-618.497
Akaike Inf. Crit.	642.467	1,082.890	372.679	741.517	946.914	716.208	1,262.994

Note:

*p<0.1; **p<0.05; ***p<0.01

.4 Values

Table .7: One-vs-All Regression Results - 5 Values

	<i>Dependent variable:</i>						
	AfD (1)	CDU (2)	CSU (3)	FDP (4)	Grüne (5)	Linke (6)	SPD (7)
Conservation	-4.277** (1.825)	5.478*** (0.933)	4.750*** (1.429)	-6.193*** (1.855)	-6.337*** (1.716)	-4.722** (1.845)	0.424 (0.852)
Openness to Change	-2.052** (0.930)	0.425 (0.738)	1.228 (1.429)	-1.496* (0.906)	5.518*** (0.887)	-2.952*** (0.912)	-0.223 (0.657)
Hedonism	4.760*** (0.959)	-1.022 (0.765)	-1.495 (1.703)	-1.088 (0.984)	0.071 (0.806)	-0.093 (0.984)	-0.296 (0.615)
Self-enhancement	-1.495** (0.704)	0.441 (0.507)	-0.106 (0.989)	1.295* (0.666)	-1.807*** (0.610)	-0.222 (0.741)	1.191*** (0.458)
Self-transcendence	-16.400*** (2.403)	0.282 (0.917)	-0.710 (1.877)	0.777 (1.299)	2.126** (1.038)	0.513 (1.355)	1.823** (0.817)
Constant	1.021** (0.446)	-2.213*** (0.399)	-4.030*** (0.767)	-1.246*** (0.463)	-4.292*** (0.495)	-0.195 (0.452)	-1.620*** (0.352)
Observations	1,141	1,141	1,141	1,141	1,141	1,141	1,141
Log Likelihood	-375.882	-536.406	-192.369	-386.273	-464.609	-359.570	-630.791
Akaike Inf. Crit.	763.764	1,084.812	396.739	784.547	941.218	731.140	1,273.581

Note:

*p<0.1; **p<0.05; ***p<0.01

Appendix C

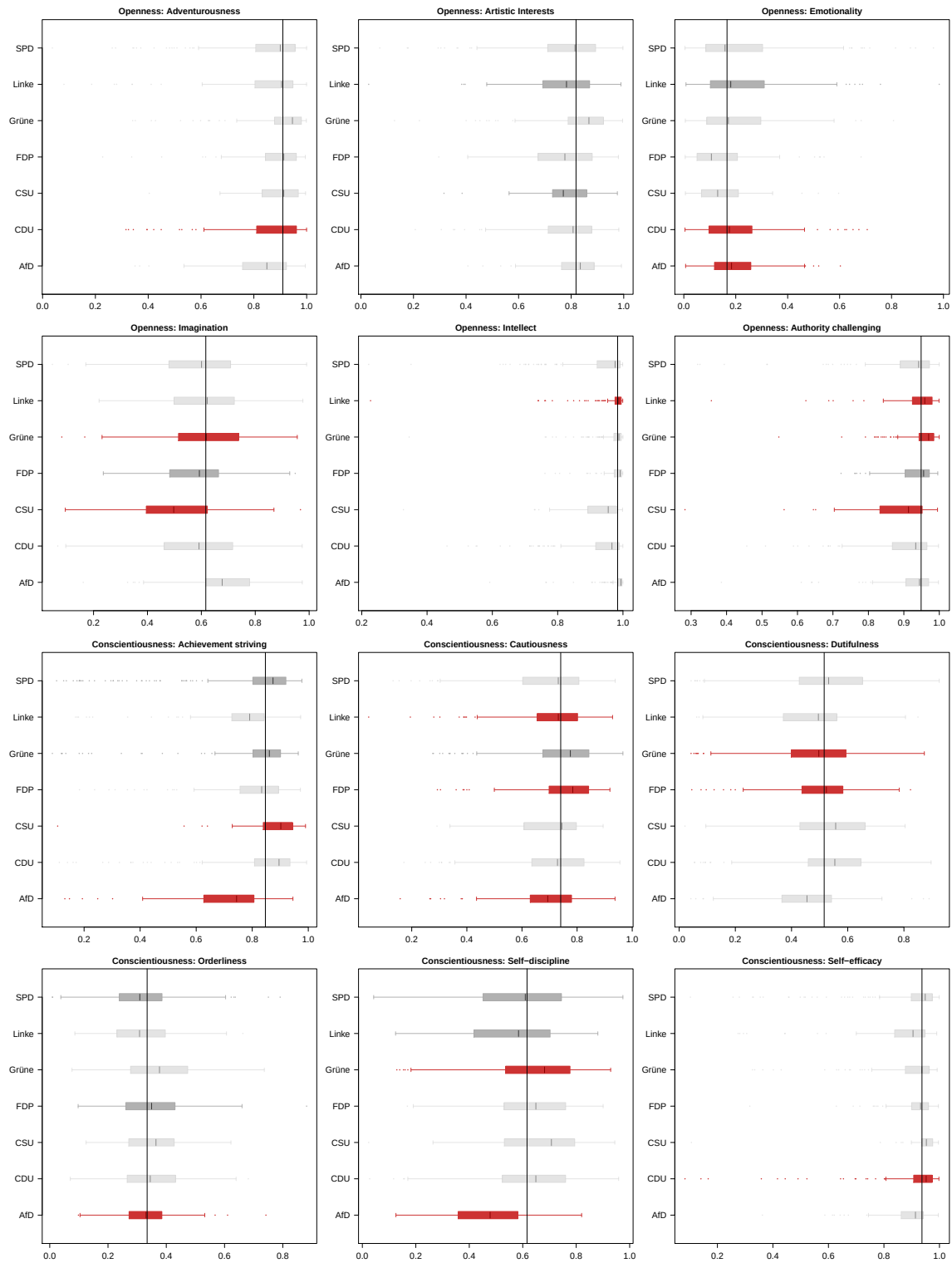


Figure .1: Distributions of scores for each party and each personality trait (30 facets) - Part 1. Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all regression.

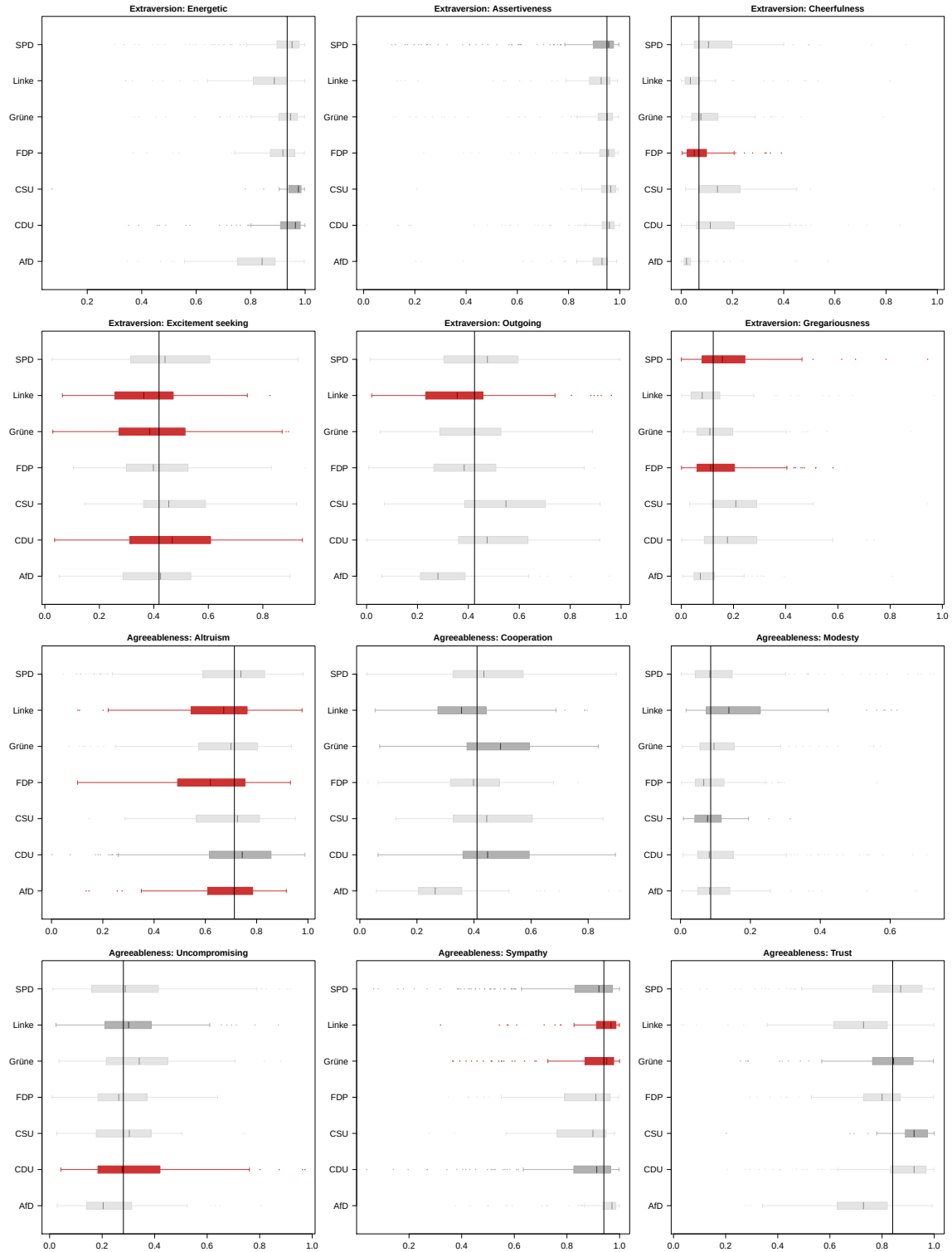


Figure .2: Distributions of scores for each party and each personality trait (30 facets) - Part 2. Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all regression.

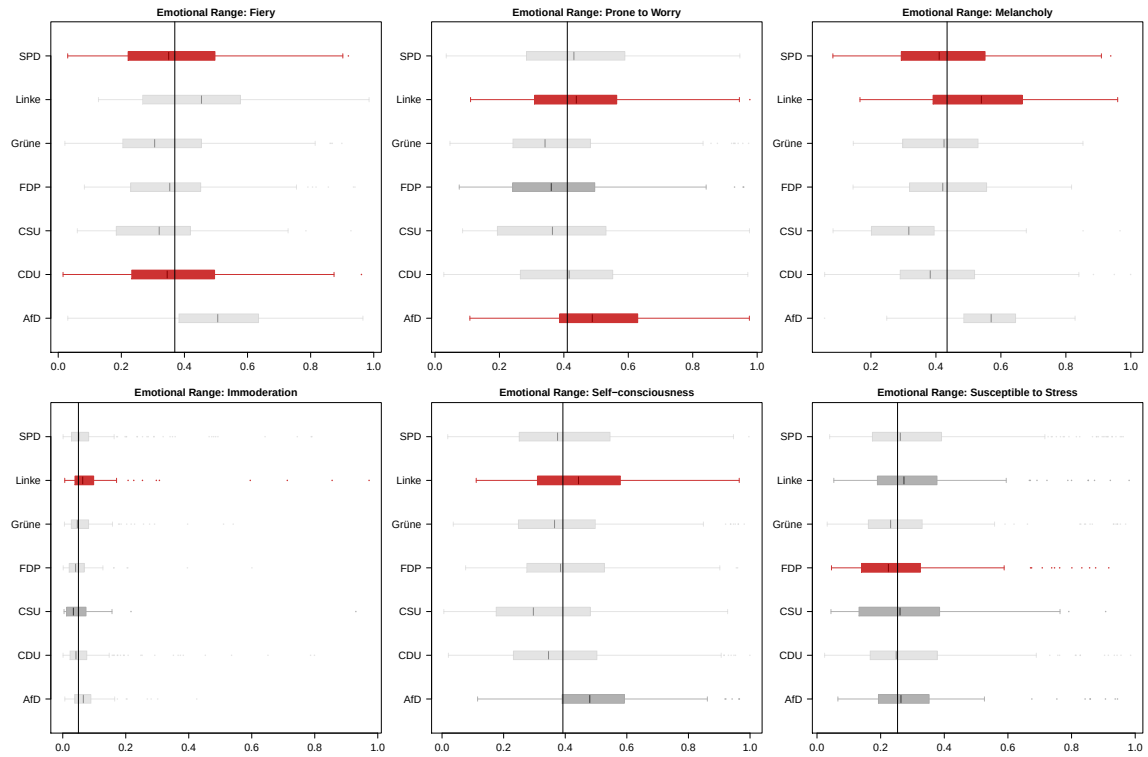


Figure .3: Distributions of scores for each party and each personality trait (30 facets) - Part 3. Light grey boxes indicate $p > 0.05$, dark grey boxes $p < 0.05$ and red boxes $p < 0.01$ of respective coefficient in the one-vs-all regression.

Appendix D

Table .8: Linear regression results (model 9-10) with general political dimensions as target variables.

	<i>Dependent variable:</i>	
	E	S
	(1)	(2)
Openness: Adventurousness	-0.275 (0.290)	0.097 (0.250)
Openness: Artistic Interests	0.914*** (0.275)	-0.784*** (0.238)
Openness: Emotionality	-0.797*** (0.281)	0.990*** (0.243)
Openness: Imagination	-0.653** (0.314)	0.872*** (0.272)
Openness: Intellect	1.295*** (0.454)	-1.061*** (0.392)
Openness: Authority challenging	-1.930*** (0.389)	1.574*** (0.336)
Conscientiousness: Achievement striving	-0.086 (0.391)	0.600* (0.338)
Conscientiousness: Cautiousness	1.932*** (0.381)	-1.728*** (0.329)
Conscientiousness: Dutifulness	0.528* (0.284)	-0.103 (0.246)
Conscientiousness: Orderliness	-0.022 (0.300)	-0.133 (0.259)
Conscientiousness: Self-discipline	-0.800** (0.328)	1.109*** (0.283)
Conscientiousness: Self-efficacy	-0.190 (0.490)	-0.133 (0.423)
Extraversion: Energetic	-0.979* (0.517)	0.976** (0.446)
Extraversion: Assertiveness	1.193*** (0.436)	-0.834** (0.377)
Extraversion: Cheerfulness	-0.689* (0.379)	-0.080 (0.327)
Extraversion: Excitement seeking	1.612*** (0.315)	-1.495*** (0.272)
Extraversion: Outgoing	-0.316 (0.340)	0.290 (0.294)
Extraversion: Gregariousness	0.711* (0.374)	0.153 (0.323)
Agreeableness: Altruism	0.600* (0.333)	-1.220*** (0.288)
Agreeableness: Cooperation	0.229 (0.342)	-0.021 (0.295)
Agreeableness: Modesty	-0.904*** (0.347)	0.745** (0.300)
Agreeableness: Uncompromising	0.797** (0.309)	-0.985*** (0.267)
Agreeableness: Sympathy	-1.216*** (0.297)	1.279*** (0.257)
Agreeableness: Trust	0.168 (0.319)	-0.282 (0.275)
Emotional Range: Fiery	1.084*** (0.331)	-0.819*** (0.286)
Emotional Range: Prone to Worry	0.104 (0.438)	-0.930** (0.378)
Emotional Range: Melancholy	0.363 (0.353)	-0.217 (0.304)
Emotional Range: Immoderation	0.296 (0.362)	-0.395 (0.313)
Emotional Range: Self-consciousness	-0.699** (0.331)	0.667** (0.286)
Emotional Range: Susceptible to Stress	0.040 (0.480)	0.787* (0.414)
Constant	-1.222* (0.713)	0.584 (0.616)
Observations	1,141	1,141
R ²	0.158	0.212
Adjusted R ²	0.135	0.190

Note: *p<0.1; **p<0.05; ***p<0.01